Timing a possible encounter

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Separate orbital period from an encounter time.

SUCCESS CRITERIA - I can...

1. Use a clearly stated period and time reference.
2. Check positions of both bodies at the same time.
3. Explain why geometry and uncertainty matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “encounter” and “period” to explain your answer. Sentence starters are optional.

encounter: objects reaching nearby positions at the same time

period: time for one complete cycle or orbit

prediction: what a model says may happen next

01 Warm-up: bring back an earlier idea

A car follows a circle at steady **5 m/s**.
1. **Sketch velocity** at two points.
2. Could net force be zero while it follows this path?

Word help: net force = the vector sum of all forces on one object

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Marzano check - circle: ① ② ③

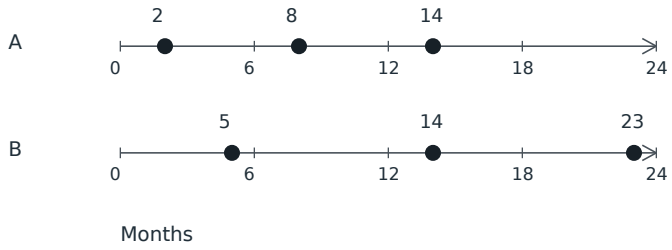
02 Worked example: explain the prediction

Reference: $t_{arrival} = t_{start}$ + travel time.
An encounter needs matching position and time. A period T is one complete orbit. Use the supplied fictional timing information.

Worked reasoning: Two objects pass the same crossing point: A at months **2**, **8**, **14** and B at months **5**, **14**, **23**. Only month **14** is a shared time in these lists.

Your turn: 1. **Draw two timelines** and connect the matching time.
2. **Explain why a shared location** alone is insufficient.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

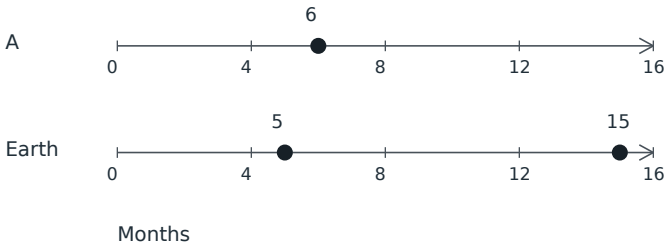


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

In a simplified orbit example, two paths cross at the same point. Object A arrives in **6 months**. Earth arrives in **5 months** and again in **15 months**. **Mark these times**. Does sharing a crossing point show that the objects will meet? Explain.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

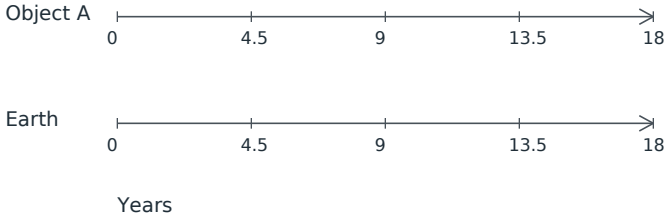


Marzano check - circle: ① ② ③

04 Apply the physics

A toy orbit repeats every **8 years**. An object was at marker P in year **0**.
1. **Name its next two returns** to P.
2. Does that tell where Earth is?

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student turns a ±**2**-day timing range into “a **50%** chance of impact.” Explain why that does not follow.

Sentence starter: I think … . I can point to …, which shows …, so … .

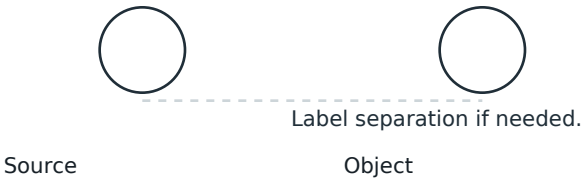
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** object is in a field of **6 N/kg**. **Give its mass** and weight, and distinguish the two quantities.
Label the masses, field or force relevant to the question.

Word help: weight = the gravitational force on an object

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: 1 2 3

07 Exit check

Name two pieces of information, beyond period, needed to assess a possible encounter.

Sentence starter: I notice … . This supports … because … .

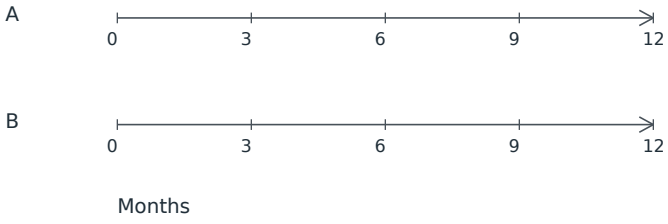
Evidence needed	What it tells us
___	___
___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Create two repeating crossing schedules that share a location but never share a time in their first **12 months**.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3